TRAIL: Trajectory-Aware Barrier Diagnostics for SGD Dynamics

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Abstract

A common way to probe neural loss landscapes is to interpolate between model states, treating endpoint geometry as a proxy for the optimization path. We show that this view can miss important structure: a linear chord can appear almost barrier-free while the realized SGD trajectory between the same checkpoints passes through elevated-loss regions. We introduce TRAIL, a trajectory-aware diagnostic framework for comparing endpoint geometry with realized optimization paths. TRAIL contrasts chord-paths from direct interpolation with observed-paths traced by intermediate training states, and summarizes their mismatch through centered loss profiles, BarrierGap, and profile-shape distances. TRAIL surfaces loss barriers that chord-based analysis misses, and we use it to map late-stage trajectory geometry across temperature regimes.

1 Introduction

Loss-landscape analysis often studies neural networks by connecting two parameter states and measuring the loss along a chosen path. Linear interpolation is especially common: if a straight chord between two checkpoints stays low-loss, the endpoints are treated as geometrically connected; if it rises, the pair is said to be separated by a barrier. This viewpoint has shaped work on visualizing loss landscapes, mode connectivity, and flatness-related generalization [Li et al., 2018, Garipov et al., 2018, Draxler et al., 2018, Fort and Jastrzebski, 2019]. Yet it answers an endpoint question. It asks whether a particular geometric shortcut is low-loss, not whether stochastic gradient descent (SGD) actually followed a low-loss route between those same states.

This distinction matters whenever optimization dynamics are themselves the object of study. SGD trajectories are shaped by the learning rate, batch size, stochastic gradient noise, BatchNorm state, and transient training history. A pair of checkpoints may be connected by an apparently benign chord even though the checkpoint sequence between them passes through elevated-loss regions. Conversely, the realized path may differ from the chord mostly by shape *below* the endpoint baseline, without producing a positive peak. Chord-only diagnostics cannot separate these cases because they discard the intermediate states actually visited by training.

We use the learning-rate to batch-size ratio $T_{\mathrm{proxy}} = \eta/B$ as an organizing axis, following work that relates SGD stochasticity to learning rate, batch size, and gradient noise [Mandt et al., 2017, Jastrzebski et al., 2018, Xie et al., 2021, Sclocchi and Wyart, 2024]. We use it only as a regime label: equal T_{proxy} need not imply equal trajectory geometry. The diagnostic question is: *when does the realized SGD trajectory contain loss-profile structure that endpoint interpolation misses?*

We first use a 16-regime grid over learning rate and batch size to show how the standard chord-barrier statistic $\Delta L = \mathrm{Peak}(\gamma_{\mathrm{chord}})$ varies with proxy temperature. We then focus on late-stage checkpoint pairs and ask a more specific question: when the endpoint chord is flat or nearly flat under a fixed evaluation protocol, does the realized SGD route have a different centered-loss profile?

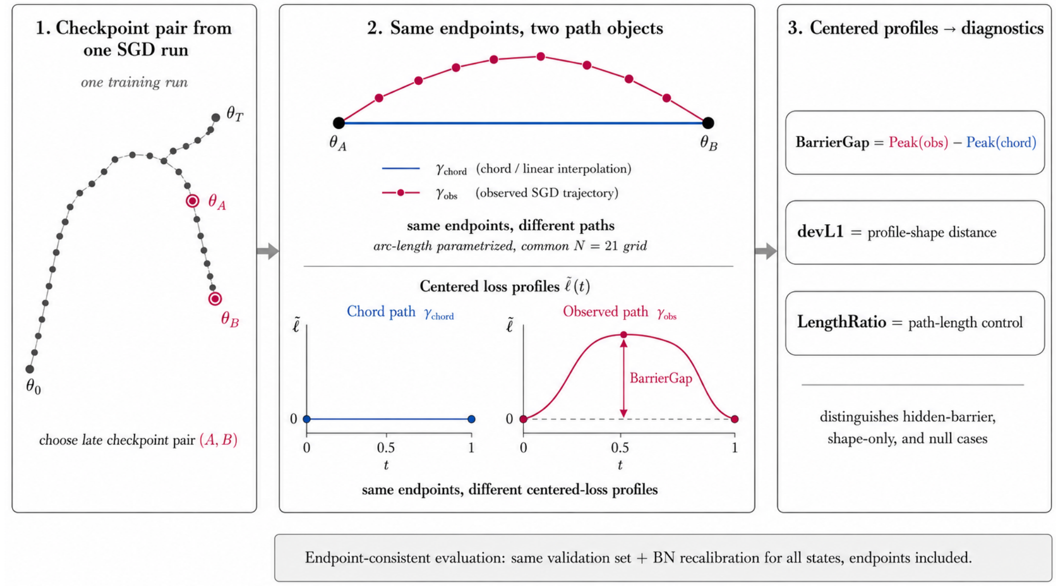

Figure 1: Conceptual overview of TRAIL. From a single SGD run, the protocol selects a checkpoint pair, compares the endpoint chord with the realized checkpoint route, and evaluates both centered-loss profiles under the same endpoint-consistent convention. The comparison isolates how two paths with identical endpoints can induce different centered-loss profiles.

We introduce TRAIL, a framework that turns this distinction into a controlled measurement. For each checkpoint pair, the *chord-path* is the direct linear interpolation between endpoints, while the *observed-path* is the checkpoint subsequence actually traversed by SGD, reparameterized by cumulative arc length. Both paths are evaluated on a common grid with the same endpoint-consistent BatchNorm recalibration protocol. Their centered-loss profiles are then compared through peak statistics, $\mathrm{BarrierGap}$, $\mathrm{devL1}$, and path-length controls.

This paper is a measurement study. Its technical contribution is the TRAIL protocol: a same-endpoint comparison of the direct chord and the realized checkpoint route, with centered profiles computed under a shared BatchNorm recalibration convention instead of mixing endpoint logs with recalibrated interior states. We apply this protocol across late-stage checkpoint pairs and find cases where flat endpoint chords coexist with elevated realized routes, alongside shape-only and null cases. On a seed-0 50→100 pair, an optimized quadratic Bezier curve remains barrier-free and a random-direction sharpness probe is much smaller than the path-level elevation, separating TRAIL from endpoint-connectivity and point-curvature explanations.

2 Related work

Loss landscapes, barriers, and mode connectivity. Empirical studies of mode connectivity challenged the picture of isolated minima separated by high barriers, showing that independently trained networks can often be connected by low-loss curves [Garipov et al., 2018, Draxler et al., 2018]. Recent work extends this idea from curves to Bezier surfaces, further emphasizing that post-hoc connectivity depends on the path family [Ren et al., 2025]. Visual and filter-normalized landscape methods made endpoint geometry easier to inspect [Li et al., 2018], while large-scale and topological views emphasized the connectedness and extended structure of low-loss regions [Fort and Jastrzebski, 2019, Akhtiamov and Thomson, 2022]. These works build paths *after the fact* between fixed endpoints. Mode connectivity therefore answers an existence question: does *some* low-loss curve connect two endpoints? TRAIL answers a different, complementary question: when the endpoints are fixed, does the path *actually traversed* by SGD exhibit barrier structure that the endpoint chord and post-hoc connectivity curves do not detect? An optimized low-loss curve and the realized SGD path are different objects: the former reflects what the loss landscape *permits*, the latter reflects what the optimizer *did*.

SGD noise, temperature, and schedules. A second line of work interprets SGD as a stochastic process and relates its behavior to learning rate, batch size, and gradient noise [Mandt et al., 2017, Jastrzebski et al., 2018, Xie et al., 2021, Sclocchi and Wyart, 2024]. Large-batch and sharp-minima studies connect optimization hyperparameters to generalization and local geometry [Keskar et al., 2017, Dinh et al., 2017, Andriushchenko et al., 2023]. Recent schedule-oriented work studies how warmup and decay shape river-valley-like loss-landscape behavior [Wen et al., 2024]. Newer analyses of trajectory sensitivity, sharpness dynamics, edge-of-stability behavior, and scalable curvature measurement provide additional context for path-level diagnostics [Kwok et al., 2025, Yoo et al., 2025, Yang et al., 2026, Xu et al., 2026, Liao et al., 2026, Kalra et al., 2026]. TRAIL complements these point- and schedule-level analyses by measuring loss profiles along checkpoint routes.

Trajectory-aware analysis. Mode connectivity has occasionally been used to interpret optimization trajectories [Gotmare et al., 2018], and weight averaging methods exploit the fact that late checkpoints may lie in a broad low-loss region [Izmailov et al., 2018]. A low-loss endpoint connection, however, does not imply that the realized path stayed low-loss: averaging or curve-finding may work because the endpoints lie in a connected basin, even if SGD moved between them through elevated regions. Our work makes this distinction explicit by evaluating the chord and observed path on the same centered-loss support under a common protocol. The relevant comparison is not only whether a low-loss connection exists in parameter space, but whether the optimizer's realized route exhibits loss-profile structure that endpoint and post-hoc connectivity diagnostics do not measure.

3 TRAIL: trajectory-aware path diagnostics

Let $\theta_0, \ldots, \theta_T$ be epoch-level checkpoints from a single training run, and let (A, B) denote a pre-specified checkpoint pair. The *chord-path* is the linear interpolation between endpoints,

$$\gamma_{\text{chord}}(t) = (1-t)\theta_A + t\theta_B, \qquad t \in [0,1]. \tag{1}$$

The *observed-path* is the piecewise-linear path through the checkpoint subsequence $(\theta_A, \theta_{A+1}, \ldots, \theta_B)$, parameterized by cumulative Euclidean arc length. The arc-length parametrization ties the path parameter to parameter displacement rather than to checkpoint index, so checkpoints with very small displacement contribute almost no arc-length mass after normalization. Both paths are evaluated on the same grid $0 = t_1 < \cdots < t_N = 1$. All reported extrema are grid-based quantities; we do not optimize continuous path extrema separately. Interpolated states are evaluation states only and are never inserted back into training.

Discrete metrics on the interpolation grid. For a path γ, define grid losses $L_i^\gamma = L(\gamma(t_i))$, endpoint baseline

$$B_i^\gamma = (1-t_i)L(\gamma(0)) + t_i L(\gamma(1)), \tag{2}$$

and centered profile $\ell_i^\gamma = L_i^\gamma - B_i^\gamma$. The endpoints lie on the baseline by construction, $\ell_1^\gamma = \ell_N^\gamma = 0$. The peak and pit are

$$\text{Peak}(\gamma) = \max_i \ell_i^\gamma, \qquad \text{Pit}(\gamma) = \min_i \ell_i^\gamma, \tag{3}$$

which are nonnegative and nonpositive respectively because the endpoints lie on the grid. For the chord-path, we write $\Delta L = \text{Peak}(\gamma_{\text{chord}})$, the standard chord-barrier statistic. Our primary discrepancy statistic is

$$\text{BarrierGap} = \text{Peak}(\gamma_{\text{obs}}) - \text{Peak}(\gamma_{\text{chord}}), \tag{4}$$

which is positive when the realized path has a higher centered-loss peak than the chord. To compare profile shape beyond the maximum, we also report the L_1 profile-shape distance

$$\text{devL1} = \frac{1}{N}\sum_{i=1}^{N} \left|\ell_i^{\text{obs}} - \ell_i^{\text{chord}}\right|. \tag{5}$$

The *LengthRatio* is the ratio of cumulative observed-path length to the endpoint chord length,

$$\text{LengthRatio} = \frac{\sum_{i=A+1}^{B} \|\theta_i - \theta_{i-1}\|_2}{\|\theta_B - \theta_A\|_2}. \tag{6}$$

By the triangle inequality, $\mathrm{LengthRatio} \geq 1$; equality holds only when the trajectory is collinear with and monotone along the chord. We use $\mathrm{LengthRatio}$ as a control rather than as a barrier metric. Full discrete definitions are reproduced in Appendix D; the evaluation protocol that fixes the family of pairs and the aggregation order is in Appendix C.

Endpoint-consistent BatchNorm recalibration. Networks with BatchNorm complicate path evaluation because interior interpolated states do not correspond to any state visited during training, and their BatchNorm running statistics are stale. The standard remedy is to recalibrate BN on a sample of training data before evaluation. Centered profiles, however, are meaningful only if endpoints *and* interior states are evaluated on the same scale. We initially evaluated endpoints from training-time logs and interior states after BN recalibration; on the anchor 70→100 pair this protocol mismatch shifted endpoint loss by 0.56-0.68 nats, an order of magnitude larger than the centered-profile signal we sought to measure, and produced spurious chord peaks. The corrected protocol resets BN statistics, recalibrates on a fixed unaugmented training sample, switches to evaluation mode, and evaluates loss for *every* path state, endpoints included. We use this endpoint-consistent convention throughout the main analysis; implementation details and checks are given in Appendices C and L.

Role of recalibration. $\mathrm{BarrierGap}$ magnitude depends on the BatchNorm recalibration convention, so recalibration is part of the diagnostic rather than a hidden preprocessing detail. Appendix I reports the recalibration sweep: the sign of the 50→100 case is stable, while the magnitude can change. For this reason, cross-slice statements emphasize prevalence, sign, and summaries within the chosen protocol; we do not treat $\mathrm{BarrierGap}$ as a normalization-independent physical scale.

What landscape does the diagnostic measure? $\mathrm{BarrierGap}$ measures elevation along a realized parameter path under a specific evaluation regime: validation loss after endpoint-consistent BN recalibration on a fixed unaugmented sample. This recalibrated centered profile is a *recalibrated evaluation landscape* for comparing on-trajectory and off-trajectory states under one BatchNorm convention; it is not the online loss stream that SGD experienced during training, where minibatch statistics and stochastic gradients change at every step. The two objects coincide at trained checkpoints only up to recalibration sample variance and generally diverge at non-checkpoint interior states. This is the same convention used in mode-connectivity and loss-landscape visualization work that places non-checkpoint states under recalibrated BN [Garipov et al., 2018, Draxler et al., 2018, Li et al., 2018].

This distinction matters for interpretation. If recalibrated diagnostic accuracy collapses at an off-trajectory state while standard validation accuracy at a trained endpoint remains high, that does not mean SGD experienced a comparable quality collapse; it means the fixed recalibrated evaluation protocol assigns high loss to that parameter state. The Bezier control is evaluated under the same convention: on the seed-0 50→100 pair, an optimized same-endpoint curve remains barrier-free while the realized checkpoint route is elevated. The reported barrier is therefore not a claim that no low-loss connection exists, nor a claim about a protocol-independent physical obstacle. It measures how the sampled SGD route scores relative to the endpoint chord under the same recalibrated evaluation rule. Cross-protocol claims, for example with a different recalibration budget, recalibration sample, or non-BN architecture, require a separate study.

4 Experimental setup

The primary setting is image classification on CIFAR-10 and CIFAR-100 with ResNet-18 and ResNet-34, trained for 100 epochs with SGD with momentum 0.9 and weight decay 5×10^{-4}. Checkpoints are saved every epoch. We evaluate late-stage pairs 50→100, 70→100, and 80→100, chosen pre-specified rather than after seeing the data. We index regimes by $T_{\mathrm{proxy}} = \eta/B$: *midT* corresponds to $\eta = 0.05, B = 128$ ($T_{\mathrm{proxy}} = 3.91 \times 10^{-4}$), *lowT* to $\eta = 0.025, B = 256$ ($T_{\mathrm{proxy}} = 9.77 \times 10^{-5}$). For the regime-grid analyses in Section 5.2 we additionally use 16 baseline regimes spanning T_{proxy} from 9.77×10^{-5} to 6.25×10^{-3} (full mapping in Appendix G, Table 7).

We aggregate the endpoint-consistent measurement set across the dataset-architecture-regime grid. Measurements computed under inconsistent endpoint baselines are used only in appendix sanity checks and excluded from the main summaries. Bootstrap confidence intervals are computed at the level of the reported claim, with the resampling unit matched to the quantity.

Dataset / arch.	Regime	Pair	Seeds	Median BG	Min-max BG	Share BG>0
CIFAR-10 / ResNet-18	midT	50→100	6	0.099	0.000-0.226	5/6
CIFAR-10 / ResNet-18	midT	70→100	6	0.048	0.000-0.234	5/6
CIFAR-10 / ResNet-18	midT	80→100	6	0.114	0.000-0.225	5/6
CIFAR-100 / ResNet-18	midT	50→100	5	1.413	0.401-2.104	5/5
CIFAR-100 / ResNet-18	midT	70→100	5	2.393	−0.319-2.699	4/5
CIFAR-100 / ResNet-18	midT	80→100	5	0.245	−0.732-2.574	4/5
CIFAR-10 / ResNet-34	midT	50→100	3	0.100	0.058-0.232	3/3
CIFAR-10 / ResNet-34	midT	70→100	3	0.174	0.066-0.297	3/3
CIFAR-10 / ResNet-34	midT	80→100	3	0.151	0.055-0.289	3/3
CIFAR-100 / ResNet-34	midT	50→100	3	0.167	0.048-0.286	3/3
CIFAR-100 / ResNet-34	midT	70→100	3	0.272	0.055-0.314	3/3
CIFAR-100 / ResNet-34	midT	80→100	3	0.268	0.239-0.362	3/3
CIFAR-10 / ResNet-18	lowT	50→100	5	0.380	−0.600-1.660	4/5
CIFAR-10 / ResNet-18	lowT	70→100	5	0.280	−0.130-0.810	3/5
CIFAR-10 / ResNet-18	lowT	80→100	5	0.000	0.000-0.140	1/5
CIFAR-10 / ResNet-18	highT	50→100	3	0.015	0.015-0.030	3/3
CIFAR-10 / ResNet-18	highT	70→100	3	0.042	0.034-0.047	3/3
CIFAR-10 / ResNet-18	highT	80→100	3	0.036	0.035-0.069	3/3

Table 1: Multi-seed endpoint-consistent evidence. Values are in native cross-entropy units and are most directly comparable within a fixed slice; cross-slice comparisons should be read through signs and shares rather than absolute scale.

5 Results

5.1 Flat endpoint chords can coexist with elevated observed paths

Table 1 reports the principal multi-seed evidence across the matched dataset/architecture slices. On the longest late window 50→100, share($\mathrm{BarrierGap} > 0$) is at least $5/6$ on every endpoint-consistent mid-temperature slice. The CIFAR-100/ResNet-18 rows contain larger within-protocol values, but the shorter windows in that slice also vary substantially across five seeds. We therefore compare magnitudes mainly within a fixed dataset, architecture, and evaluation convention, and use cross-slice summaries primarily for prevalence and sign.

5.2 Temperature-modulated regime structure

The auxiliary 16-regime grid studies the standard chord-barrier statistic $\Delta L = \mathrm{Peak}(\gamma_{\mathrm{chord}})$. Hotter $T_{\mathrm{proxy}} = \eta/B$ regimes show larger and more frequent chord barriers across threshold sweeps (Appendix G). The main TRAIL question is stricter: among late-stage pairs with flat or nearly flat chords, does the realized path still show elevated centered loss?

On CIFAR-10/ResNet-18, the lowT-midT-highT triptych gives three distinct geometries. At lowT, positive $\mathrm{BarrierGap}$ concentrates on the broad 50→100 window and largely disappears on 80→100. At midT, 14/15 late-stage measurements are positive with the largest magnitudes in this slice. At highT, all nine measured late-stage pairs are positive but small. Thus T_{proxy} organizes the observations, but it does not determine a unique path geometry or a monotone $\mathrm{BarrierGap}$ law.

5.3 Related baselines: mode-connectivity curve and point sharpness

We also check two standard alternatives on the CIFAR-10/ResNet-18 midT seed-0 50→100 pair. A quadratic Bezier curve with one optimized control point remains barrier-free under the same endpoint-consistent evaluation: both the chord and Bezier path have centered peak 0.000, while the observed path reaches +0.1149. At radius $r = 0.1$, the observed peak is only modestly sharper than endpoints or the chord midpoint (mean loss change +0.0029 versus +0.0017-+0.0019). Table 2 summarizes the comparison; broader curve families and spectral sharpness estimates are left for future work.

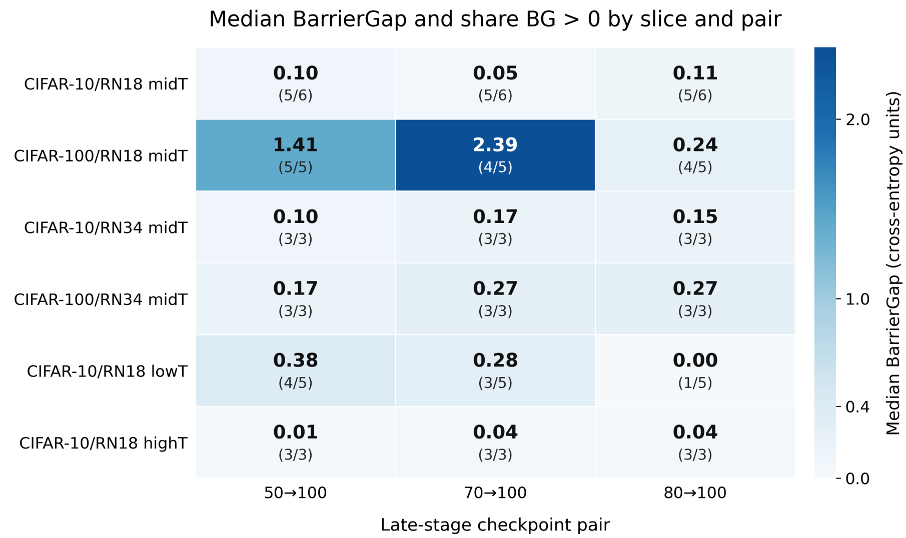

Figure 2: Median $\mathrm{BarrierGap}$ and share of positive measurements by slice and late-stage pair. Mid-temperature slices are consistently positive; the lowT-midT-highT rows show that prevalence and magnitude are temperature-modulated rather than determined by T_{proxy} alone.

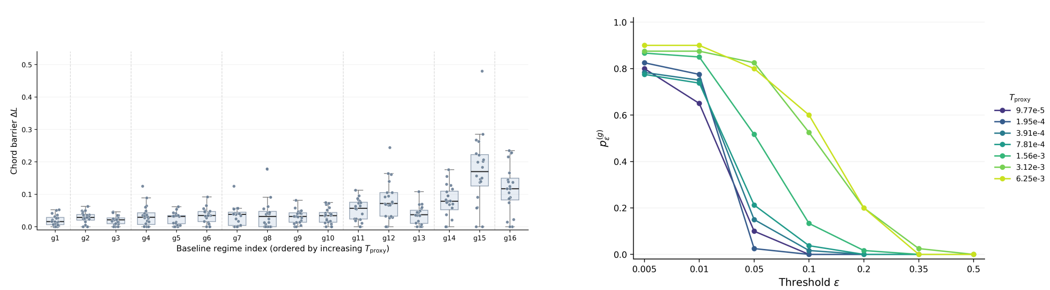

Figure 3: Auxiliary regime-grid context. Left: chord-barrier distributions across the 16 learning-rate/batch-size regimes. Right: threshold frequencies for chord barriers aggregated by shared T_{proxy}. This figure contextualizes the temperature organization of endpoint geometry; the main TRAIL discrepancy analysis is reported separately in Table 1 and Figure 2.

5.4 Protocol and nuisance checks

Table 3 summarizes the main protocol checks on the seed-0 50→100 case. Grid refinement leaves the reported peak unchanged, so the signal is not a discretization artifact. The BN-recalibration sweep is more delicate: the sign remains positive, but the magnitude varies substantially, so we use it as a protocol caveat rather than an invariance result. Replacing the train-log endpoint baseline with the recalibrated baseline removes inflated chord peaks from the inconsistent baseline, and arc-length and checkpoint-index parametrizations of the observed path agree to four decimal places. Subsampling checkpoints at step ≤ 2 preserves the signal; at step 5 it collapses, indicating that the sampled observed path requires sufficient checkpoint resolution.

5.5 Additional validity checks

Simple geometric quantities are relevant but do not fully explain the diagnostic. Across endpoint-consistent measurements, $\mathrm{BarrierGap}$ has weak Spearman correlation with $\mathrm{LengthRatio}$ ($\rho = 0.15$); $\mathrm{devL1}$ has essentially no rank correlation with $\mathrm{LengthRatio}$ ($\rho = -0.03$). $\mathrm{BarrierGap}$ also has weak rank correlation with the recalibrated endpoint-loss gap ($\rho = 0.15$). $\mathrm{devL1}$, by contrast, correlates more strongly with endpoint-loss gap ($\rho = 0.70$), so we treat it as a profile-shape measure with a scale caveat rather than as a standalone barrier statistic.

Check	Object	Quantity	Value	Interpretation
Chord path	linear interpolation	centered peak	0.0000	barrier-free endpoint chord
Bezier path	optimized quadratic curve	centered peak	0.0000	low-loss connection exists
Observed path (TRAIL)	realized SGD route	centered peak	+0.1149	realized route is elevated
Endpoint sharpness	$\theta_{50}, \theta_{100}, r = 0.1$	mean ΔL	+0.0017-+0.0018	small point-level change
Chord-mid sharpness	$t = 0.5, r = 0.1$	mean ΔL	+0.0019	small point-level change
Observed-peak sharpness	$t = 0.45, r = 0.1$	mean ΔL	+0.0029	modestly sharper, not explanatory

Table 2: Same-pair comparison against related diagnostics on the seed-0 50→100 pair. The optimized Bezier curve tests post-hoc mode connectivity; the sharpness rows test whether the path-level peak is reproduced by local random-direction curvature. Full per-grid and per-radius values are in Appendix F.

Check	Setting	50→100 BG	Interpretation	Claim use
BN-recalibration sweep	budgets varied	0.006 / 0.011 / 0.226	sign stable; magnitude varies strongly	caveat
Grid-resolution sweep	grid refined	0.226 / 0.226 / 0.226	peak is grid-stable	support
Endpoint fix	trainlog vs. recal endpoints	inconsistent-baseline peaks removed	removes baseline-mismatch artifact	support
Identity / two-point sanity	observed=chord	zero residual	implementation sanity	support
Parametrization	arc vs. index	agree to 4 decimals	not a sensitivity axis	support

Table 3: Main protocol checks for the seed-0 50→100 case. Full recalibration, grid-resolution, and path-length controls are reported in Appendices I and J.

In the length-only specification $\mathrm{BarrierGap} = \beta_0 + \beta_1\,\mathrm{LengthRatio} + \xi$, the coefficient is $+0.0101$ ($p < 10^{-3}$, $R^2 = 0.288$). Adding T_{proxy} and window width raises R^2 only to 0.292, and neither additional control is significant. Length, proxy temperature, and window width account for part of the variation, but they do not determine the row-level values. The residual variation should be read conservatively: it may include slice effects, checkpoint sampling, and evaluation-protocol sensitivity. Positive $\mathrm{BarrierGap}$ remains common within $\mathrm{LengthRatio}$ bins (0.62 in Q1 to 0.96 in Q4), and windows with the same width still differ appreciably (Appendix J).

Table 4 summarizes these checks. TRAIL does not produce positive barriers everywhere: the measurements include hidden-barrier, shape-only, null, and intermediate cases.

6 Discussion

TRAIL sits between several familiar diagnostics. Endpoint interpolation and mode-connectivity curves ask whether a low-loss connection can be constructed between two states; on the seed-0 50→100 pair, the optimized Bezier curve gives such a connection even though the sampled SGD route is elevated. Sharpness probes local curvature near a state; the random-direction probe is consistent with the observed peak being slightly sharper, but its scale is far smaller than the centered path elevation. Proxy temperature helps organize the regimes, but lowT, midT, and highT differ in both prevalence and magnitude, so equal η/B should not be read as fixing the trajectory.

BatchNorm is the main interpretive caveat: the reported profiles are recalibrated validation losses under one endpoint-consistent convention, not the online minibatch losses optimized by SGD. Whether the same qualitative pattern persists with normalization schemes that do not use running statistics remains an empirical question.

7 Limitations

TRAIL is a measurement framework, and the empirical setting here is kept controlled: supervised image-classification trajectories, SGD training, and a BatchNorm recalibration convention that evaluates the two path objects on the same scale. This choice makes the comparisons clean, but it also leaves broader architectures, other optimizers, and normalization schemes without running statistics for future work.

The main interpretive caveat is that centered profiles are recalibrated validation losses, not the online minibatch losses seen during training. Recalibration is used because it is the standard way to evaluate non-checkpoint path states and because it lets chord, Bezier, and observed paths be compared under

Validity question	Result	Consequence
$\mathrm{BarrierGap}$ vs. path length	Spearman $+0.15$	not fully explained by $\mathrm{LengthRatio}$
$\mathrm{BarrierGap}$ vs. endpoint gap	Spearman $+0.15$	not fully explained by endpoint drift
$\mathrm{devL1}$ vs. endpoint gap	Spearman $+0.70$	shape metric has scale caveat
Length+temperature+width regression	$R^2 = 0.288 \to 0.292$	simple controls leave residual variation
Case taxonomy	3 hidden, 3 shape-only, 4 null, 17 intermediate	not all-positive
Standard vs. recal. accuracy	can diverge by 78-79 pp	diagnostic loss is not final quality

Table 4: Additional validity checks. "Hidden" denotes hidden observed-path barriers; "shape-only" denotes $\mathrm{BarrierGap} \leq 0.01$ with nonzero profile discrepancy. The standard-vs-recal divergence is interpreted as a protocol caveat and detailed in Appendix K.

one convention. The magnitude of $\mathrm{BarrierGap}$ should therefore be read within that convention. Our Bezier and sharpness checks cover one same-pair comparison; extending them to richer curve families, spectral curvature estimates, and models without BatchNorm is left for future work.

8 Conclusion

TRAIL reframes loss-landscape diagnostics from endpoint connectivity alone to the comparison of endpoint and realized-route profiles under a fixed evaluation protocol. Across the measured slices, cases with a flat chord and elevated observed path recur, while shape-only and null cases show that the diagnostic is not uniformly positive. The Bezier and sharpness controls illustrate why realized-route measurement is distinct from post-hoc low-loss connectivity and point-level curvature. Extending the analysis to broader model families and stronger connectivity or curvature baselines is left for future work.

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A Per-seed BarrierGap on the main slices

The aggregate measurements in Table 1 use the runs available for each slice, so the seed count is not the same everywhere. On CIFAR-10/ResNet-18 midT, the three late windows have five positive rows out of six each: the 50→100 values are $0.226, 0.151, 0.099, 0.098, 0.071, 0.000$, the 70→100 values range from 0.000 to 0.234, and the 80→100 values range from 0.000 to 0.225. The lowT slice is less uniform: the broad 50→100 window remains mostly positive, while 80→100 is mostly null. The highT slice is positive over the measured three seeds, but the magnitudes are small in all three late windows.

The CIFAR-100/ResNet-18 midT slice contains larger observed magnitudes and also substantial seed-to-seed variation. Its 50→100 rows are all positive $(2.104, 2.043, 1.413, 0.865, 0.401)$, while the shorter windows include both large positive values and negative rows: for example, 70→100 contains -0.319 alongside values above 2, and 80→100 contains -0.732 alongside a maximum of 2.574. For this reason, the main text uses this slice as evidence that large within-protocol values can occur, not as a stable cross-dataset effect-size estimate.

B Anchor centered-loss profiles

Figure 4 gives the single-run profile visualization used for the hidden-barrier example. It is kept in the appendix because the main evidence is multi-seed and multi-slice.

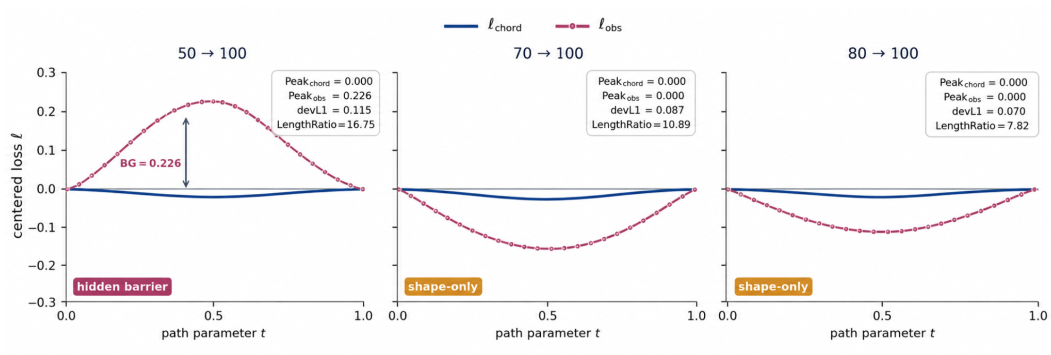


Figure 4: Single-run centered-loss profiles on the seed-0 CIFAR-10/ResNet-18 trajectory. The 50→100 window is a hidden-barrier case, while the shorter windows are shape-only cases under the centered-profile taxonomy.

C Evaluation protocol for path-based diagnostics

Each training run produces an epoch-level checkpoint sequence $\theta_0, \theta_1, \dots, \theta_T$. Path-based diagnostics are evaluated on a fixed family of checkpoint pairs (A, B) with $0 \le A < B \le T$, drawn from a pre-specified milestone schedule spanning early, middle, and late training. The same family of pairs is used for all regimes. Late-stage analyses further restrict attention to long-window milestone pairs whose endpoints both lie beyond the main optimization transient.

For a given pair (A, B), the chord-path is the linear interpolation between θ_A and θ_B. The observed-path is built from the checkpoint subsequence $(\theta_A, \theta_{A+1}, \dots, \theta_B)$ using cumulative Euclidean arc-length normalization. The path parameter is therefore tied to parameter displacement rather than checkpoint index. Checkpoints with very small displacement contribute almost no arc-length mass after normalization. All pathwise quantities are evaluated on a common interpolation grid $0 = t_1 < t_2 < \cdots < t_N = 1$, shared by the chord-path and the observed-path for the same pair. Reported extrema are therefore grid-based quantities.

Interpolated and finite-difference states are evaluation states only: they are never inserted back into training. This includes $\gamma_{\text{chord}}(t_i)$, $\gamma_{\text{obs}}(t_i)$, and perturbation states $\theta \pm \varepsilon u$. Validation loss at these states is measured after recalibration of BatchNorm running statistics on non-augmented training data. Recalibration updates BatchNorm buffers only; trainable parameters are left unchanged.

For each checkpoint pair (A, B), we record $\mathrm{Peak}(\gamma_{\text{chord}}), \mathrm{Pit}(\gamma_{\text{chord}}), \mathrm{Peak}(\gamma_{\text{obs}}), \mathrm{Pit}(\gamma_{\text{obs}})$, ΔL, $\mathrm{BarrierGap}$, $\mathrm{devL1}$, and $\mathrm{LengthRatio}$. All of these quantities are computed at the pair level. Aggregation across regimes, stages, or intervention conditions is performed only afterward. When multiple runs are available for the same dataset-architecture-regime configuration, run identity is preserved in intermediate summaries. Bootstrap confidence intervals are computed at the level of the reported quantity. Late-stage discrepancy is defined by the pre-specified late milestone family rather than by a retrospective loss threshold or epoch cutoff.

D Discrete metric definitions on the interpolation grid

All path-based quantities are evaluated on the common interpolation grid $0 = t_1 < t_2 < \cdots < t_N = 1$, shared by the chord-path and the observed-path for a given checkpoint pair (A, B). For any path $\gamma : [0, 1] \to \mathbb{R}^P$, define the discrete loss profile, endpoint baseline, and centered profile by

$$L_i^\gamma = L(\gamma(t_i)), \qquad B_i^\gamma = (1 - t_i)L(\gamma(0)) + t_i L(\gamma(1)), \qquad \ell_i^\gamma = L_i^\gamma - B_i^\gamma, \qquad i = 1, \dots, N.$$

For the two path objects of interest, $L_i^{\text{chord}} = L(\gamma_{\text{chord}}(t_i))$, $L_i^{\text{obs}} = L(\gamma_{\text{obs}}(t_i))$, $\ell_i^{\text{chord}} = \ell_i^{\gamma_{\text{chord}}}$, and $\ell_i^{\text{obs}} = \ell_i^{\gamma_{\text{obs}}}$. In exact arithmetic $\ell_1^{\gamma} = \ell_N^{\gamma} = 0$, since the profile is centered relative to the linear baseline between endpoint losses.

Define the discrete peak and pit by $\text{Peak}_N(\gamma) = \max_i \ell_i^{\gamma}$ and $\text{Pit}_N(\gamma) = \min_i \ell_i^{\gamma}$. Because the endpoints belong to the grid, $\text{Peak}_N(\gamma) \geq 0$ and $\text{Pit}_N(\gamma) \leq 0$. For the chord-path, the barrier statistic is

$$\Delta L_N = \text{Peak}_N(\gamma_{\text{chord}}) = \max_{1\leq i\leq N} \left[L_i^{\text{chord}} - \big((1-t_i)L(\theta_A) + t_i L(\theta_B)\big) \right].$$

We write $\Delta L = \Delta L_N$. The primary discrepancy statistic is $\text{BarrierGap}_N = \text{Peak}_N(\gamma_{\text{obs}}) - \text{Peak}_N(\gamma_{\text{chord}})$, and we drop the subscript N throughout. To compare full centered profiles, $\text{devL1}_N = N^{-1} \sum_i |\ell_i^{\text{obs}} - \ell_i^{\text{chord}}|$.

For a threshold $\varepsilon > 0$, the chord-barrier indicator is $\mathbf{1}_{\varepsilon}^{\text{chord}} = \mathbf{1}\{\Delta L > \varepsilon\}$. In threshold-sensitivity analyses this indicator is evaluated over $\mathcal{E} = \{0.005, 0.01, 0.05, 0.1, 0.2, 0.35, 0.5\}$; for an aggregation unit containing M pairs the corresponding exceedance frequency is $p_{\varepsilon} = M^{-1} \sum_m \mathbf{1}\{\Delta L_m > \varepsilon\}$.

For a checkpoint pair (A, B), the observed-path length and endpoint chord length are $L_{\text{path}}(A, B) = \sum_{i=A+1}^{B} \|\theta_i - \theta_{i-1}\|_2$ and $L_{\text{chord}}(A, B) = \|\theta_B - \theta_A\|_2$, giving $\text{LengthRatio} = L_{\text{path}}(A, B)/L_{\text{chord}}(A, B)$. Whenever $L_{\text{chord}}(A, B) > 0$, the triangle inequality implies $\text{LengthRatio} \geq 1$; equality holds only when the sampled trajectory is collinear with the chord and monotone along it. LengthRatio is used only as a control.

All reported quantities are grid-based path diagnostics. We do not extrapolate them to continuous-path surrogates. This keeps the chord and observed paths on common support and avoids separate numerical optimization of pathwise extrema.

E Training and intervention configurations

Each training run produces a checkpoint sequence $\theta_0, \dots, \theta_T$ with $T = 100$. Checkpoints are saved once per epoch, so the sampled training trajectory is $\Theta = (\theta_0, \dots, \theta_T)$. For each run, evaluation pairs are drawn from a fixed milestone set $\mathcal{M} = \{m_1, \dots, m_K\}$ with $0 \leq m_1 < \cdots < m_K \leq T$; pair construction is restricted to ordered milestone pairs $(A, B) \in \mathcal{P}$ with $A, B \in \mathcal{M}$ and $A < B$. Late-stage analyses further restrict attention to pairs with $A, B \in \mathcal{M}_{\text{late}} \subset \mathcal{M}$.

For an original checkpoint θ_t, validation loss $L(\theta_t)$ is evaluated directly. For any non-checkpoint state $\widetilde{\theta}$ obtained by interpolation or finite-difference perturbation, validation loss is evaluated after BatchNorm recalibration on non-augmented training data, with budget R_{BN}. All pathwise quantities are evaluated on the common grid with $N = 21$. Bootstrap confidence intervals are computed with the resampling unit matched to the reported quantity: pair-level summaries are bootstrapped over evaluated checkpoint pairs, run-level summaries over runs.

Intervention experiments modify the learning-rate schedule within an otherwise fixed baseline run. Given an intervention window $[e_{\text{start}}, e_{\text{end}}]$ with $1 \leq e_{\text{start}} \leq e_{\text{end}} \leq T$ and a multiplier $m > 0$, the intervened schedule is

$$\eta_{\text{int}}(e) = \begin{cases} m\, \eta_{\text{base}}(e), & e \in [e_{\text{start}}, e_{\text{end}}], \\ \eta_{\text{base}}(e), & \text{otherwise.} \end{cases}$$

Local geometry is evaluated at checkpoints θ_{pre}, θ_{post}, and θ_{final}, corresponding respectively to the state immediately before the intervention window, immediately after it, and at the end of training. The finite-difference curvature proxy is $\kappa(\theta; u) = [L(\theta + \varepsilon u) + L(\theta - \varepsilon u) - 2L(\theta)]/\varepsilon^2$ with $\varepsilon = \alpha\|\theta\|_2$, and its directional average over M random unit directions is $\bar{\kappa}(\theta) = M^{-1} \sum_m \kappa(\theta; u_m)$. We use $\alpha = 10^{-3}$ and $M = 100$. Coverage of the downstream analyses is uneven across dataset-architecture pairs; the main regime-level and late-stage analyses are supported most completely for CIFAR-10/ResNet-18.

F Bezier and sharpness baseline details

This appendix records the same-pair baselines used in Section 5.3. All values use the CIFAR-10/ResNet-18 midT, seed-0, 50→100 pair, the common grid $N = 21$, $R_{\text{BN}} = 50$, and the recalibrated endpoint baseline. The Bezier baseline optimizes one quadratic control point initialized at the chord midpoint for 10 passes over the t-grid; BatchNorm buffers are frozen during control-point optimization and reset/recalibrated during evaluation.

Path	Centered peak	Centered pit	Gap vs. observed
Linear chord	0.0000	-0.0104	+0.1149
Optimized quadratic Bezier	0.0000	-0.0478	+0.1149
Observed SGD path	+0.1149	-0.1176	-

Table 5: Bezier baseline on the seed-0 50→100 pair. Both the chord and optimized Bezier curve are barrier-free under the centered profile, while the realized path is elevated.

State probed at radius $r = 0.1$	Mean ΔL	Std.	Min	Max
θ_{50} endpoint	+0.001733	0.002080	-0.002003	+0.005286
θ_{100} endpoint	+0.001765	0.003803	-0.004557	+0.006858
Chord midpoint ($t = 0.5$)	+0.001868	0.002961	-0.002778	+0.005549
Observed peak ($t = 0.45$)	+0.002936	0.000758	+0.001393	+0.003970

Table 6: Random-direction sharpness probe on the seed-0 50→100 pair. The observed peak is modestly sharper but the absolute curvature gap is much smaller than the path-level centered elevation.

G Regime grid and chord-barrier scaling

For a baseline regime $g = (\eta, B)$, let $\mathcal{P}_g$ denote the set of evaluated checkpoint pairs collected from all runs assigned to that regime. For any pair-level statistic $S_{A,B}^{(g)}$, define the regime-level mean $\overline{S}^{(g)} = |\mathcal{P}_g|^{-1} \sum_{(A,B)\in\mathcal{P}_g} S_{A,B}^{(g)}$. Across the baseline regime grid, the chord-barrier statistic ΔL shifts upward with increasing $T_{\text{proxy}} = \eta/B$. The shift is more evident in the pair-level distributions than in a strictly monotone regime-by-regime ordering; the corresponding distribution and threshold plots are shown in Figure 3 in the main text.

Regime	η	B	T_{proxy}
g_1	0.025	256	9.77×10^{-5}
g_2	0.025	128	1.95×10^{-4}
g_3	0.05	256	1.95×10^{-4}
g_4	0.025	64	3.91×10^{-4}
g_5	0.05	128	3.91×10^{-4}
g_6	0.1	256	3.91×10^{-4}
g_7	0.025	32	7.81×10^{-4}
g_8	0.05	64	7.81×10^{-4}
g_9	0.1	128	7.81×10^{-4}
g_{10}	0.2	256	7.81×10^{-4}
g_{11}	0.05	32	1.56×10^{-3}
g_{12}	0.1	64	1.56×10^{-3}
g_{13}	0.2	128	1.56×10^{-3}
g_{14}	0.1	32	3.12×10^{-3}
g_{15}	0.2	64	3.12×10^{-3}
g_{16}	0.2	32	6.25×10^{-3}

Table 7: Regime mapping for the 16-regime chord-barrier grid.

η	B	T_{proxy}	BarrierGap	min-max	devL1	min-max
0.025	64	3.91×10^{-4}	0.122	0.110-0.130	0.063	0.058-0.073
0.05	64	7.81×10^{-4}	0.180	0.155-0.193	0.078	0.072-0.088
0.1	64	1.56×10^{-3}	0.237	0.208-0.261	0.111	0.098-0.138
0.2	64	3.12×10^{-3}	0.203	0.157-0.287	0.127	0.115-0.140

Table 8: Late-pair discrepancy across four CIFAR-10/ResNet-18 baseline regimes. Each row aggregates the available members of the pre-specified late-stage family for one regime.

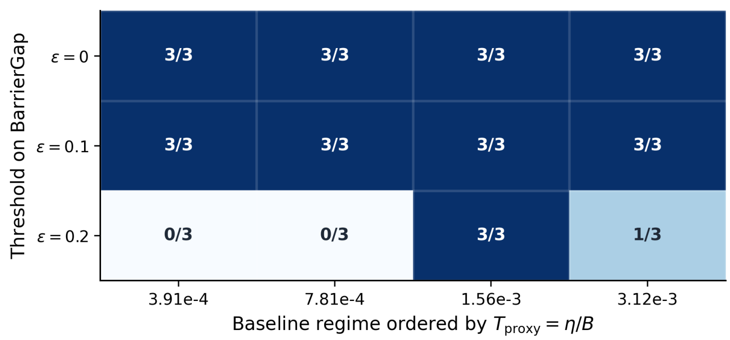


Figure 5: Frequency of late-stage pairs with $\mathrm{BarrierGap} > \varepsilon$ across baseline regimes ordered by T_{proxy}. Every cell at $\varepsilon = 0$ and $\varepsilon = 0.1$ contains 3/3 positive pairs.

To check that this pattern is not tied to one cutoff, define $p_\varepsilon^{(g)} = |\mathcal{P}_g|^{-1} \sum_{(A,B)\in\mathcal{P}_g} \mathbf{1}\{\Delta L_{A,B}^{(g)} > \varepsilon\}$ for $\varepsilon \in \mathcal{E}$. The threshold sweep in Figure 3 shows that hotter regimes retain larger barrier-event frequencies across low and moderate thresholds. The separation weakens only at the largest thresholds, where extreme chord barriers are rare for all regimes.

Late-stage discrepancy is summarized on the available members of the pre-specified late-stage family, $50 \to 100$, $70 \to 100$, $80 \to 100$. These long-window pairs isolate internal path discrepancy after endpoint losses have become close. Table 8 reports $\mathrm{BarrierGap}$ and $\mathrm{devL1}$ for four baseline regimes spanning the temperature range.

At the regime level, hotter baselines exhibit larger chord barriers in both the distributional summaries and the thresholded frequencies. Positive late-stage discrepancy also remains visible after endpoint losses become close, including in the hotter part of the grid.

H Extended late-stage discrepancy analyses

Let $\mathcal{P}_{\text{late}} = \{(50,100),(70,100),(80,100)\}$. For a baseline regime g, define $\mathcal{P}_{g,\text{late}} = \mathcal{P}_g \cap \mathcal{P}_{\text{late}}$. For each late-stage pair $(A,B) \in \mathcal{P}_{g,\text{late}}$, we report $\Delta L_{A,B}^{(g)} = \mathrm{Peak}(\gamma_{\text{chord}}^{A,B,g})$, $\mathrm{BarrierGap}_{A,B}^{(g)} = \mathrm{Peak}(\gamma_{\text{obs}}^{A,B,g}) - \mathrm{Peak}(\gamma_{\text{chord}}^{A,B,g})$, and $\mathrm{devL1}_{A,B}^{(g)} = N^{-1}\sum_i |\ell_{\text{obs}}^{A,B,g}(t_i) - \ell_{\text{chord}}^{A,B,g}(t_i)|$.

A thresholded late-stage frequency is $\pi_{g,\text{late}}(\varepsilon) = |\mathcal{P}_{g,\text{late}}|^{-1} \sum_{(A,B)\in\mathcal{P}_{g,\text{late}}} \mathbf{1}\{\mathrm{BarrierGap}_{A,B}^{(g)} > \varepsilon\}$, $\varepsilon \geq 0$. Figure 5 shows that for the evaluated baseline regimes, every late-stage pair satisfies $\mathrm{BarrierGap} > 0$; the same remains true at $\varepsilon = 0.1$; at $\varepsilon = 0.2$ no late pairs exceed the threshold in the two colder regimes, all three late pairs exceed it at $T_{\text{proxy}} = 1.56 \times 10^{-3}$, and one of the three late pairs exceeds it at $T_{\text{proxy}} = 3.12 \times 10^{-3}$.

Table 9 summarizes the three late windows separately.

Window	mean BarrierGap	min-max	mean devL1	share$(\mathrm{BarrierGap} > 0)$
$50 \to 100$	0.213	0.126-0.287	0.089	1.00
$70 \to 100$	0.173	0.155-0.208	0.086	1.00
$80 \to 100$	0.171	0.158-0.261	0.110	1.00

Table 9: Window-specific late-stage summary over the baseline regimes included in this appendix.

R_{BN}	Median ΔL	IQR ΔL	Median $\lvert\Delta L - \Delta L^{(100)}\rvert$	Flip rate ($\Delta L \geq 0.05$)
5	0.0435	0.0178-0.0588	0.0120	25.0%
20	0.0500	0.0250-0.0750	0.0035	8.3%
50	0.0525	0.0270-0.0815	0.0010	0.0%
100	0.0535	0.0278-0.0833	0.0000	0.0%

Table 10: Effect of BatchNorm recalibration budget on the chord-barrier statistic ΔL for the 24-pair baseline subset. Most of the variation is concentrated at the smallest recalibration budgets; by $R_{\mathrm{BN}} = 50$ the summaries are close to the $R_{\mathrm{BN}} = 100$ reference.

N	Mean proxy ΔL bound	Share ($\Delta L > 0$) bound	Share ($\Delta L \geq 0.05$) bound	Peak-support bound
11	[0.0213, 0.0221]	[8.1%, 8.4%]	[6.6%, 6.9%]	[99.7%, 100.0%]
21	0.0221	8.4%	6.9%	100.0%
41	≥ 0.0221	$\geq 8.4\%$	$\geq 6.9\%$	[99.4%, 100.0%]

Table 11: Restricted interpolation-grid check around the observed $N = 21$ setting. Reported quantities are based on the proxy `baseline_max_diff`; this table is not a direct multi-N rerun.

For $\tau_{\mathrm{flat}} \geq 0$, define the flat-chord late-stage subset $\mathcal{P}^{\mathrm{flat}}_{g,\mathrm{late}} = \{(A, B) \in \mathcal{P}_{g,\mathrm{late}} : \Delta L^{(g)}_{A,B} \leq \tau_{\mathrm{flat}}\}$. With $\tau_{\mathrm{flat}} = 0$, this restriction selects all 12 evaluated late-stage pairs from the regimes considered here. On this subset, the median $\mathrm{BarrierGap}$ is 0.179, and the share of pairs with positive $\mathrm{BarrierGap}$ is 1. The late-stage discrepancy does not disappear when endpoint losses become close: it is present in every evaluated member of the late family, remains present at threshold $\varepsilon = 0.1$, and persists under the flat-chord restriction $\tau_{\mathrm{flat}} = 0$.

I Robustness to recalibration, resolution, and geometry settings

This appendix reports three evaluation-protocol checks: BatchNorm recalibration, interpolation-grid resolution, and the finite-difference geometry probe. The recalibration check is the most direct; the grid-resolution check is narrower, and the geometry probe records the finite-difference configuration used in the experiments.

For BatchNorm recalibration, Table 10 reports the recalibration sweep for a 24-pair baseline subset. The median chord-barrier statistic ΔL increases from 0.0435 at $R_{\mathrm{BN}} = 5$ to 0.0535 at $R_{\mathrm{BN}} = 100$, while the median deviation from the $R_{\mathrm{BN}} = 100$ reference decreases from 0.0120 to 0. The threshold-instability rate at $\Delta L \geq 0.05$ falls from 25.0% at $R_{\mathrm{BN}} = 5$ to 0% at $R_{\mathrm{BN}} = 50$ and $R_{\mathrm{BN}} = 100$. Most variation is concentrated at the smallest recalibration budgets; by $R_{\mathrm{BN}} = 50$ the ΔL summaries are already close to the $R_{\mathrm{BN}} = 100$ reference, justifying $R_{\mathrm{BN}} = 50$ as the chosen recalibration convention (Section 3).

Table 11 reports a restricted resolution check around the operating grid $N = 21$. The $N = 21$ row is the stored 21-point result, the $N = 11$ row is a conservative bounded estimate derived from the same summary, and the $N = 41$ row is a one-sided lower bound. A direct rerun on the same seed-0 50→100 pair gives $\mathrm{BarrierGap} = 0.226$ for $N \in \{11, 21, 41\}$ at $R_{\mathrm{BN}} = 50$, supporting grid stability under the chosen protocol.

For the finite-difference geometry probe, $\kappa(\theta; u) = [L(\theta + \varepsilon u) + L(\theta - \varepsilon u) - 2L(\theta)]/\varepsilon^2$ with $\varepsilon = \alpha\|\theta\|_2$, and $\bar{\kappa}(\theta) = M^{-1} \sum_m \kappa(\theta; u_m)$. The reported geometry results use $\alpha = 10^{-3}$ and $M = 100$. The repository preserves this default finite-difference configuration but does not contain a standalone same-family sensitivity sweep over alternative values of α or M.

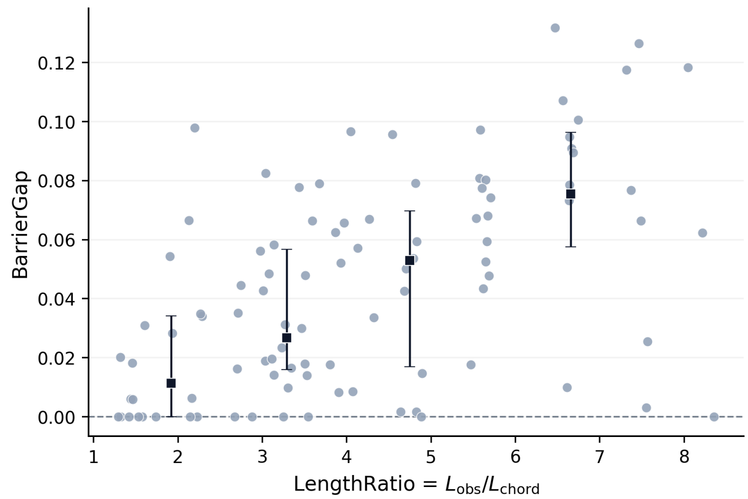


Figure 6: $\mathrm{BarrierGap}$ versus $\mathrm{LengthRatio}$ in the baseline control block. Points denote evaluated pairs. Square markers and error bars denote binned summaries.

LengthRatio bin	median BarrierGap	share $(\mathrm{BarrierGap} > 0)$	median devL1
Q1	0.011	0.62	0.028
Q2	0.027	0.88	0.032
Q3	0.053	0.96	0.036
Q4	0.075	0.96	0.046

Table 12: Summary within $\mathrm{LengthRatio}$ bins in the baseline control block. Both $\mathrm{BarrierGap}$ and $\mathrm{devL1}$ increase across bins on average, but each bin still contains visible variation.

Among the three protocol checks, the recalibration evidence is the strongest. The resolution evidence is narrower but does not support a large hidden chord-side discretization effect around $N = 21$. For geometry, the appendix records the finite-difference setting used for the reported curvature values: $\alpha = 10^{-3}$ and $M = 100$. We do not include a separate sweep over alternative probe settings.

J Path-length and sampling controls

This appendix evaluates whether the observed-chord discrepancy can be reduced to path length or to simple sampling effects induced by wider checkpoint windows. Let $\mathrm{LengthRatio}_{A,B} = \sum_{i=A+1}^{B} \|\theta_i - \theta_{i-1}\|_2 / \|\theta_B - \theta_A\|_2$ and $W_{A,B} = B - A$.

Figure 6 shows the relation between $\mathrm{BarrierGap}$ and $\mathrm{LengthRatio}$ in the baseline control block. Larger $\mathrm{LengthRatio}$ is associated with larger $\mathrm{BarrierGap}$ on average; the residual spread within bins remains substantial, so path length alone does not determine the discrepancy.

Table 12 summarizes the same relation within $\mathrm{LengthRatio}$ bins. Both $\mathrm{BarrierGap}$ and $\mathrm{devL1}$ increase across the bins, and the fraction of pairs with positive $\mathrm{BarrierGap}$ also rises.

A regression control gives the same conclusion. In the length-only specification $\mathrm{BarrierGap} = \beta_0 + \beta_1 \,\mathrm{LengthRatio} + \xi$, the estimated coefficient of $\mathrm{LengthRatio}$ is $+0.0101$ with $p < 10^{-3}$ and $R^2 = 0.288$. After adding T_{proxy}, the fit changes only slightly to $R^2 = 0.291$, and T_{proxy} is not significant. After adding both T_{proxy} and W, the fit changes again only slightly, to $R^2 = 0.292$. Across all reported specifications, $\mathrm{LengthRatio}$ remains positive and significant, whereas the additional controls are not significant.

Window width provides a second control for sampling structure. Since the observed-path for a wider pair contains more intermediate checkpoints, a purely mechanical explanation would predict larger discrepancy at larger W. Table 13 reports the corresponding window-level summaries. Wider

Window	W	mean $\mathrm{BarrierGap}$
$5 \to 10$	5	0.023
$10 \to 15$	5	0.034
$15 \to 20$	5	0.036
$20 \to 30$	10	0.064
$30 \to 40$	10	0.063
$40 \to 50$	10	0.046

Table 13: Window-level control summary in the baseline control block.

windows tend to exhibit larger discrepancy on average, but width alone does not remove the variation: windows with the same width still differ appreciably in $\mathrm{BarrierGap}$, and the rise with W is not strictly monotone across the evaluated windows.

Both $\mathrm{LengthRatio}$ and window width are associated with larger discrepancy, but neither removes the effect. Positive $\mathrm{BarrierGap}$ remains common within comparable path-length bins, and simple controls based on T_{proxy} or W leave substantial residual variation.

K Intervention coverage and regime maps

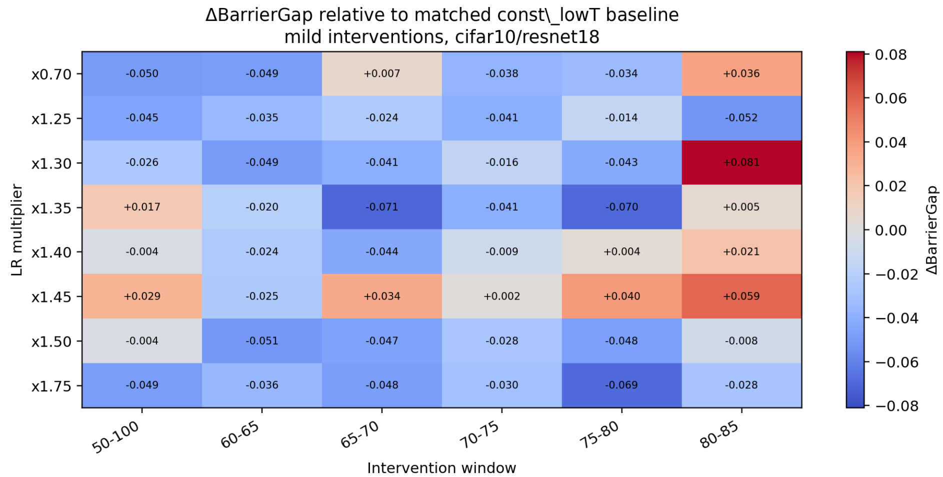
This appendix reports the intervention measurements used as diagnostic checks. We include three groups of measurements: a mild-intervention grid, a small long-window block, and a late matched-control comparison. Final-quality comparisons are available across the measured dataset-architecture slices. Matched discrepancy summaries are available only for ResNet-18 slices, and local-geometry measurements only for CIFAR-10/ResNet-18. Accordingly, the visual summaries are restricted to CIFAR-10/ResNet-18.

The intervention measurements are heterogeneous. Local heating does not induce a uniform shift in discrepancy statistics across windows and amplitudes. On the most complete slice, some interventions yield positive matched discrepancy gain at small final-quality cost, whereas others reduce final quality. Across the broader set of runs, final-quality comparisons are available more widely than matched discrepancy or geometry summaries, so the intervention-level discussion remains concentrated in the CIFAR-10/ResNet-18 block.

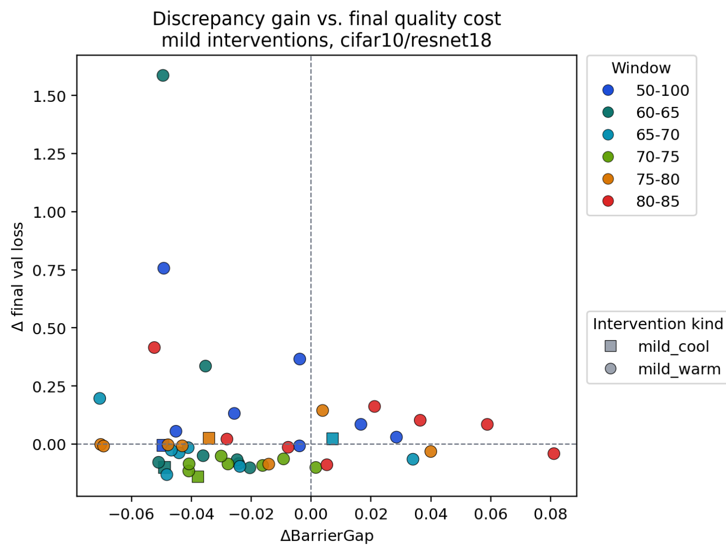
Diagnostic and standard accuracy can diverge. We also include a branching run to compare standard validation accuracy with BN-recalibrated diagnostic accuracy. This comparison is used only to show that the two quantities are distinct under this protocol.

Branch	LR after 50	Standard acc.	Recal. acc.	Drop	BG
baseline	0.050	0.812	0.545	-27 pp	0.226
drop	0.025	0.886	0.100	-79 pp	0.304
drop	0.010	0.883	0.100	-78 pp	-0.097
hot	0.100	0.819	0.826	+1 pp	0.000

Table 14: Standard validation accuracy and BN-recalibrated diagnostic accuracy in the reported intervention slice. The divergence is used as an interpretation caveat for recalibrated diagnostics, not as evidence for a training controller.



(a) Matched discrepancy map



(b) Quality-discrepancy trade-off

Figure 7: Intervention coverage on CIFAR-10/ResNet-18. The matched-discrepancy map is shown at larger scale because its cell annotations are the information-bearing part of the figure. The lower panel summarizes the corresponding final-quality trade-off.

L Endpoint-consistency sanity checks

The implementation sanity suite includes identity-path checks, two-point observed-equals-chord checks, synthetic two-dimensional profiles, an endpoint-consistency test, and post-fix regression tests for the CLI. The identity and two-point tests ensure that the path math returns zero residual when the observed path is constructed to equal the chord.

Pair / protocol	$\text{Peak}_{\text{chord}}$	Peak_{obs}	BG	Interpretation
50→100 inconsistent fast	1.62	1.62	0.002	inflated common baseline
70→100 inconsistent confirm	0.65	0.59	-0.054	artifact inverse signal
50→100 endpoint-consistent	0.00	0.226	0.226	hidden barrier example
70→100 endpoint-consistent	0.00	0.00	0.000	shape-only / below baseline

Table 15: Inconsistent-baseline versus endpoint-consistent diagnostics. Inconsistent-baseline references document the protocol correction and are not used for the main claims.

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